

D. Inversion Value of a Permutation

time limit per test: 3 seconds
 memory limit per test: 512 megabytes

A permutation of length n is an array of n integers, where each number from 1 to n appears exactly once. An *inversion* in a permutation p is a pair of indices (i, j) such that $i < j$ and $p_i > p_j$.

For a permutation p , we define its *inversion value* as the number of its subsegments that contain at least one inversion. Formally, this is the number of pairs of integers (l, r) ($1 \leq l < r \leq n$) for which there exists a pair of indices (i, j) satisfying the following conditions: $l \leq i < j \leq r$ and $p_i > p_j$.

For example, for the permutation $[3, 1, 4, 2]$, the inversion value is 5.

You are given two integers n and k . Your task is to construct a permutation of length n with an **inversion value equal to exactly k** .

Input

The first line contains one integer t ($1 \leq t \leq 500$) — the number of test cases.

Each test case consists of a single line containing two integers n and k ($2 \leq n \leq 30$;

$$0 \leq k \leq \frac{n(n-1)}{2}).$$

Output

For each test case, output the answer as follows:

- if the desired permutation does not exist, output a single integer 0;
- otherwise, output n distinct integers from 1 to n — the desired permutation. If there are multiple such permutations, you may output any of them.

Example

input	Copy
5 4 5 5 10 5 0 6 8 3 1	
output	Copy
3 1 4 2 5 4 3 2 1 1 2 3 4 5 2 3 5 6 1 4 0	

Educational Codeforces Round 183 (Rated for Div. 2)

Contest is running

00:35:31

Contestant



→ Submit?

Language: Python 3.13.2
 Almost always, if you send a solution on PyPy, it works much faster

Choose file: Choose File No file chosen

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